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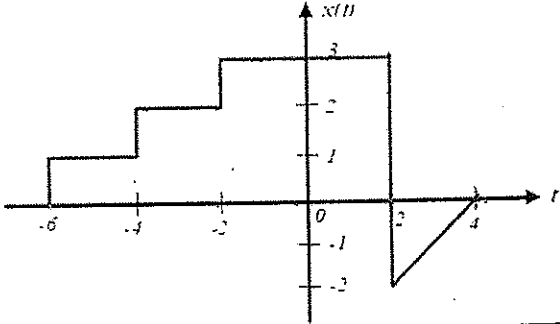
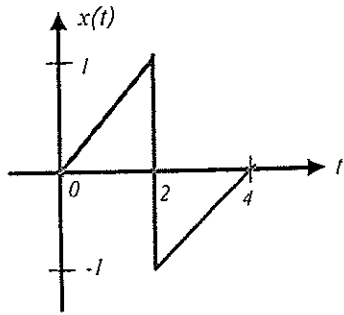
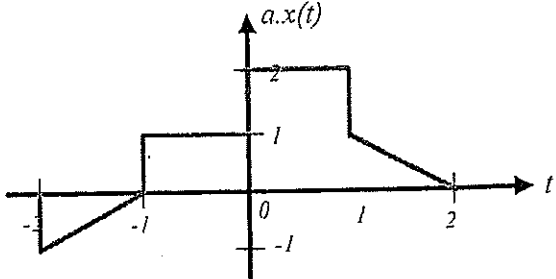
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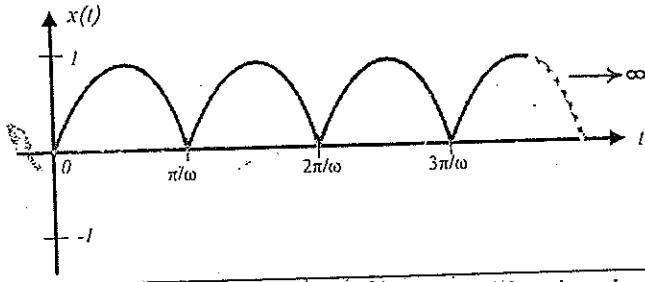
2nd SEMESTER B. Tech. ICE & EE
MID-SEMESTER EXAMINATION, JUNE 2022

Course Code: ICICC03/EEEEEC03
Time: 1.5 Hours

Course Title: Signals and Systems
Max. Marks: 25

Note: - Attempt all questions. Missing data/ information if any, maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	COs
1a	The signal $x(t)$ is shown below. Apply the signal transformation operations to draw the signal $x(-3t+9)$. 	2.5	CO1
1b	Express the following signal $x(t)$ in equation form as a sum of elementary signals: 	2.5	CO1
2	Given a signal $a.x(t)$ as below, 		
2a	Construct the signal, $y_1(t) = x(t-1)$.	2.5	CO1
2b	Construct the signal, $y_2(t) = [x(t) + x(-t)]u(t)$	2.5	CO1
3	Given a system, $y[n] = x^2[n] + 1/x^2[n-1]$		

3a	Determine whether the system is Linear or Non-Linear.	2.5	CO1
3b	Determine whether the system is Time-invariant or Time-variant.	2.5	CO1
4a	Prove that the product of two even or odd signals is an even signal; and the product of even signal and odd signal is an odd signal.	2.5	CO2
4b	Prove that the signals $x_1(t) = e^{-at}u(t)$ and $x_2(t) = -e^{-at}u(-t)$ have the same laplace transform and differ only in ROC.	2.5	CO2
5a	Find the laplace transform of a periodic waveform $x(t)$ starting from $t = 0$ sec., as shown below, 	2.5	CO2
5b	Find the laplace transform of $x(t) = te^{-2t}\sin 2t.u(t)$ using the properties of laplace transform.	2.5	CO2
